CY201	Cyber Security Principles and Concepts (3 CH)	Spring 2026	CYS

Pre-Requisite: N/A
Instructor: **Muhammad Ahmad Nawaz**
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Office Hours: 11:00am ~ 2:00 pm

Course Introduction

This program will offer a comprehensive exploration of foundational cybersecurity principles and practical skills essential for safeguarding digital assets. Delivered through a dynamic curriculum, best practices, and real-world case studies, the course covers critical topics including network security, web security, operating system security, mobile security, cloud security, and emerging trends.

Course Contents

Students will explore Cyber Security Principles and Concepts, encompassing areas like Network, Web, Mobile, cloud security, , and emerging trends. Through theoretical study with exercises, students will gain vital skills to adeptly navigate and secure modern digital environments.

Mapping of CLOs and GAs

Sr. No	Course Learning Outcomes+	GAs*	Bloom's Taxonomy level (Cognitive domain)
CLO-1	Understand the fundamental principles of cybersecurity.	GA 2 (Knowledge for Solving Computing Problems)	C2 (Understand)
CLO-2	Understand the cyber security conceptual knowledge to analyze and implement security measures.	GA 2 (Knowledge for Solving Computing Problems)	C2 (Understand)
CLO-3	Apply a profound comprehension of cybersecurity principles and concepts.	GA3 (Problem Analysis)	C3 (Apply)
	+Please add the prefix "Upon successful completion of this course, the student will be able to"		

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes	30%	20%	20%
Assignments/Project	5%	20%	20%
Midterm Exam	35%	30%	30%
Final Exam	30%	30%	30%

Overall Grading Policy

Assessment Items	Percentage
Quizzes	10%
Project	10%
Assignments	10%
Midterm Exam	30%
Final Exam	40%

Text and Reference Books

Textbooks:

- W. Stallings, *"Network Security Essentials: Applications and Standards"*, 7th Edition, Boston, Massachusetts, United States of America, Pearson Education, April 2025.ISBN: 9780138201548
- R. S. Sandhu and R. J. B. Boles, *"Cybersecurity: Principles and Practice"*, 3rd Edition, New York, United States of America, Wiley, March 2025.ISBN: 9781119791004
- J. Andress, *"The Basics of Information Security"*, 3rd Edition, Boston, Massachusetts, United States of America, Syngress (Elsevier), August 2024.ISBN: 9780128129951

Reference book:

- M. T. Simpson, K. Backman, and J. E. Corley, "The Art of Vulnerability Assessment and Penetration Testing", 1st Edition, Boca Raton, Florida, United States of America, Auerbach Publications, June 3, 2019.
ISBN: 9781138339996
- T. L. Wilson, R. C. Newman, and S. S. Le Blond, "Cybersecurity and Applied Mathematics: A Hands-On Approach", 1st Edition, Boca Raton, Florida, United States of America, CRC Press, February 2025.
ISBN: 9780367758914

09/01/26

- C. P. Pfleeger, S. L. Pfleeger, and J. Margulies, "Security in Computing", 6th Edition, Upper Saddle River, New Jersey, United States of America, Pearson Education, January 2025.
  ISBN: 9780134744781
- B. Schneier, "Applied Cryptography: Protocols, Algorithms, and Source Code in C", 25th Anniversary Edition, 2nd Edition, Hoboken, New Jersey, United States of America, Wiley, May 2025.
  ISBN: 9781119096726
- K. Scarfone and P. Mell, "Guide to Network Security", 2nd Edition, Gaithersburg, Maryland, United States of America, NIST Press, October 2025.ISBN: 9780160934802
- E. Cole, S. D. Ring, and J. Morell, "Handbook of Information Security, Threats, Vulnerabilities, Prevention, Detection and Management", 2nd Edition, Boca Raton, Florida, United States of America, CRC Press, July 2025.
  ISBN: 9780367758952

Administrative Instruction

- According to institute policy, 80% attendance is *mandatory* to appear in the final examination.
- Assignments must be submitted as per instructions mentioned in the assignments.
- In any case, there will be no retake of (scheduled/surprise) quizzes.
- For queries, kindly follow the office hours to avoid any inconvenience.

Computer Usage/Software Tool

- The course will incorporate practical computer usage and software tools for hands-on application, reinforcing cybersecurity principles through simulations, analysis, and implementation exercises aligned with each week's topics.

Lecture Breakdown

Week 1	**Introduction to Cyber Security** • Overview of Cyber Security • Cyber Threats and Attack Vectors • CIA Triad
Week 2	**Cryptography and Encryption, Network Security** • Privacy Concepts • Principles of Cryptography • Public Key Infrastructure (PKI), Cryptographic Attacks • Network Basics Protocols, OSI model, and basic networking concepts • Firewalls and IDPS • Virtual Private Networks (VPNs)
Week 3	**Social Engineering** • Definition, Types (Phishing, Vishing, Tailgating, Dumpster Diving, Shoulder Surfing, Eavesdropping), Public Hotspot Attacks, QR code Attacks
Week 4	**Web Security** • Web Application Security • Secure Coding Principles • Browser Security and HTTPS
Week 5	**Operating System Security, Mobile Security** • Securing Windows, Linux, macOS • Access Controls and Permissions • Patch Management • Mobile Device Security • Mobile Application Security, Mobile Device Management (MDM)
Week 6	**Digital Forensics** • Introduction to Digital Forensics, History, Standards, Fundamentals of Digital Forensic Investigations
Week 7	**Cloud Security** • Cloud Computing Security Basics • Securing Cloud Infrastructure • Best practices for securing cloud environments. • Shared Responsibility Model
Week 8	**Risk Management and Incident Handling** • Definition of Risk, Risk Management Process, Incident Response, Disaster Recovery, Business Continuity, Physical Security
Week 9	**Identity and Access Management** • IAM Fundamentals, Identity, and access management principles

	• Authentication Methods (Multi-factor authentication, biometrics, etc). • Authorization and Access Control
Week 10	**Security for IoT and Embedded Systems** • IoT Security Challenges (Risks and vulnerabilities in IoT) • Securing Embedded Systems Case Studies and Practical Examples
Week 11	**Application Security, Security for IoT and Embedded Systems** • IoT Security Challenges (Risks and vulnerabilities in IoT) • Securing Embedded Systems • Secure Software Development Lifecycle
Week 12	**Cyber Threat Intelligence** • Threat Intelligence Fundamentals • Understanding threat intelligence • Threat Feeds and Analysis • Utilizing threat feeds and analyzing threats • Incident Attribution and Reporting
Week 13	**Security Policies and Compliance** • Developing Security Policies • Creating effective security policies • Regulatory Compliance • Understanding and adhering to regulations. • Security Audits and Assessments
Week 14	**Crafting a Cyber Security Policy** • Challenges (Risks Importance and Design Cycle of a Cyber Security Policy, Policy Benchmarks, Example Policies
Week 15	**Case Study and Project Presentation** • Project Vivas and demonstration • Report or Conference Paper Writing

Version	Spring 2026
Name of Instructor	Engr. Muhammad Ahmad Nawaz
Instructor's Signature	
Date	09/1/26
Name of HoD	Prof. Dr. Ghulam Abbas
HoD's Signature	09/01/26
Date	January 7, 2026